Adding (1) and (2), $3x - 2y + z = 2$, $2x + 3y - z = 5$ (+)

$5x + y = 7$ …(4)

Adding (2) and (3), $2x + 3y - z = 5$, $x + y + z = 6$ (+)

$3x + 4y = 11$ …(5)

$4 \times (4) - (5)$ $20x + 4y = 28$, $3x + 4y = 11$ (−)

$17x = 17 \Rightarrow x = 1$

Substituting $x = 1$ in (4), $5 + y = 7 \Rightarrow y = 2$

Substituting $x = 1$, $y = 2$ in (3), $1 + 2 + z = 6$ we get, $z = 3$

Therefore, $x = 1$, $y = 2$, $z = 3$

Example 3.4 In an interschool atheletic meet, with total of 24 individual prices, securing a total of 56 points, a first place secures 5 points, a second place secures 3 points, and a third place secures 1 point. Having as many third place finishers as first and second place finishers, find how many athletes finished in each place.

Solution Let the number of I, II and III place finishers be x, y and z respectively.

Total number of events $= 24$; Total number of points $= 56$.

Hence, the linear equations in three variables are

$x + y + z = 24$ …(1) $5x + 3y + z = 56$ …(2) $x + y = z$ …(3)

Substituting (3) in (1) we get, $z + z = 24 \Rightarrow z = 12$

$\therefore$ (3) will be, $x + y = 12$

(2) $\Rightarrow$ $5x + 3y = 44$

$3 \times$ (3) $\Rightarrow$ $3x + 3y = 36$ (−)

$2x = 8$ we get, $x = 4$

Substituting $x = 4$, $z = 12$ in (3) we get, $y = 12 - 4 = 8$

Therefore, Number of first place finishers is 4

Number of second place finishers is 8

Number of third place finishers is 12.

Example 3.5 Solve $x + 2y - z = 5$; $x - y + z = -2$; $-5x - 4y + z = -11$

Solution $x + 2y - z = 5$ …(1) $x - y + z = -2$ …(2) $-5x - 4y + z = -11$ …(3)

Adding (1) and (2) we get, $x + 2y - z = 5$, $x - y + z = -2$ (+)

$2x + y = 3$ …(4)

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